

ORANGE COUNTY AP, NY, USA

WMO: 725015

Lat: 41.509N

Lon: 74.265W

Elev: 365

StdP: 14.50

Time Zone: -5.00 (NAE)

Period: 99-19

WBAN: 04789

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	2.7	8.4	-8.7	3.5	5.9	-4.3	4.4	11.6	24.1	23.4	20.9	25.9	4.6	300	0.488

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	21.8	90.6	73.7	88.0	72.6	84.9	70.9	76.4	86.0	75.0	83.8	73.5	81.6	9.0	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
73.4	125.7	80.8	72.3	121.1	79.6	71.0	115.9	78.2	40.0	86.1	38.6	83.7	37.3	81.7	82.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
19.9	17.6	15.8	DB	-4.4	95.8	6.7	4.5	-9.2	99.0	-13.1	101.6	-16.8	104.2	-21.7	107.4	
			WB	-4.8	79.2	6.4	1.8	-9.4	80.5	-13.2	81.6	-16.8	82.6	-21.5	83.9	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	50.3	26.3	28.6	37.3	49.1	59.4	68.0	72.8	71.1	63.9	52.2	41.6	31.9
	DBStd	18.02	10.20	9.30	9.24	8.50	7.55	6.12	5.18	5.28	6.88	7.92	8.12	8.64
	HDD50	2773	736	600	406	115	8	0	0	0	1	69	274	564
	HDD65	6057	1200	1019	858	482	210	39	4	10	103	403	703	1026
	CDD50	2879	1	1	13	88	299	539	706	654	418	138	20	2
	CDD65	690	0	0	0	5	36	128	245	199	70	7	0	0
	CDH74	6531	0	0	3	125	465	1224	2326	1704	619	65	0	0
	CDH80	2147	0	0	0	36	139	375	868	550	170	9	0	0
Wind	WSAvg	5.9	6.9	7.0	7.4	7.2	5.7	5.3	4.8	4.3	4.5	5.7	6.0	6.5
Precipitation	PrecAvg	43.7	3.0	2.4	3.4	3.7	4.2	4.3	3.9	3.8	4.0	3.5	3.7	3.2
	PrecMax	58.9	9.8	4.6	7.6	8.8	10.2	11.5	13.6	9.6	9.9	17.9	7.6	7.5
	PrecMin	28.3	0.2	0.2	0.4	1.2	1.2	0.4	0.8	0.6	0.7	0.0	0.8	0.8
	PrecStd	7.6	1.8	1.0	1.5	1.5	2.1	2.4	2.3	2.0	2.4	2.8	1.7	1.6
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	58.3	60.5	70.9	85.1	88.9	91.2	95.0	94.6	90.1	80.6	69.9	61.9
		MCWB	53.3	53.4	56.4	63.0	70.1	74.3	75.5	75.2	73.5	67.1	60.6	56.0
	2%	DB	50.3	52.7	63.9	76.9	84.1	87.9	91.2	89.5	84.8	74.8	64.1	54.6
		MCWB	45.4	45.4	52.8	59.3	68.1	73.3	74.4	74.0	70.7	64.0	57.2	50.2
	5%	DB	44.4	46.1	57.4	71.1	79.5	84.2	88.1	85.7	81.1	70.5	60.4	49.3
		MCWB	39.8	39.8	47.2	56.6	65.7	70.5	73.1	71.8	68.7	62.7	53.2	44.7
	10%	DB	40.0	41.7	52.3	65.4	74.5	81.4	84.7	82.6	77.2	66.3	55.3	45.0
		MCWB	35.6	36.2	44.9	53.0	62.6	69.1	71.5	70.1	67.2	59.1	49.3	40.8
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	55.2	56.0	59.3	66.4	73.9	76.9	78.9	78.4	76.2	70.8	63.1	57.9
		MCDB	57.3	59.2	66.8	79.1	84.3	86.5	90.6	89.0	84.5	75.9	66.9	60.3
	2%	WB	46.1	46.4	54.1	62.4	70.6	74.9	76.6	76.0	73.7	67.3	58.8	51.0
		MCDB	49.2	52.1	61.7	73.1	80.2	84.2	86.7	84.7	81.2	72.9	63.2	54.1
	5%	WB	40.3	40.7	49.4	58.8	67.8	73.1	75.2	74.3	71.3	63.9	54.5	45.8
		MCDB	43.6	45.7	55.9	68.5	76.3	81.4	84.2	82.5	77.8	69.3	58.2	48.6
	10%	WB	36.6	36.5	45.4	54.6	65.1	71.0	73.5	72.5	69.3	60.2	50.6	41.0
		MCDB	39.7	40.9	51.5	62.7	72.1	78.5	81.5	79.8	75.1	65.3	54.6	44.5
Mean Daily Temperature Range	5% DB	MDBR	16.7	18.4	19.1	22.5	22.4	21.4	21.8	21.2	22.2	20.9	19.3	15.4
		MCDBR	20.8	24.4	28.2	32.6	28.9	25.7	25.1	24.5	25.9	26.5	25.3	20.5
		MCWBR	17.0	18.4	17.9	17.3	15.3	12.4	10.7	10.8	13.4	16.3	18.1	17.0
	5% WB	MCDBR	20.2	23.1	25.9	28.7	24.6	22.5	21.4	20.6	21.0	22.4	21.7	18.7
		MCWBR	17.4	18.4	18.0	17.2	14.2	11.9	10.1	10.2	11.9	15.8	17.7	17.0
Clear-Sky Solar Irradiance	taub	0.300	0.321	0.347	0.363	0.406	0.467	0.474	0.455	0.383	0.355	0.320	0.302	
	taud	2.414	2.369	2.384	2.391	2.296	2.160	2.152	2.197	2.374	2.461	2.514	2.495	
	Ebn at Noon	270	281	286	288	276	259	255	257	270	266	262	261	
	Edh at Noon	25	31	34	36	41	47	47	44	34	28	23	22	
All-Sky Solar Radiation	RadAvg	559	836	1147	1458	1629	1768	1840	1609	1334	894	620	456	
	RadStd	42	82	89	155	164	167	103	104	145	98	60	48	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (59 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+196	+110

Nomenclature: See separate page